

4.4 Solving Determinants by Hand



Core skills you'll master in this section:

- ▶ The minor matrix: remove the row and column containing the specified element.
- ▶ The minor: the determinant of the minor matrix, a scalar.
- ▶ The cofactor: a sign adjustment applied to the minor.
- ▶ Laplace expansion: pick any row or column, multiply each element by its cofactor, and sum the results to get the determinant.
- ▶ Preferentially expanding along the row or column with the most zeros reduces the amount of computation.

In this section, we'll use the 3×3 determinant as an example to introduce how to solve determinants by hand.

Readers who don't need to understand how to solve determinants by hand can skip this section.

Although earlier in this chapter we already gave the formula for the determinant of the 3×3 matrix $\mathbf{A}$:

$$\det(\mathbf{A}) = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32}$$

However, this formula isn't easy to remember. So, we'll introduce a more structured method – Laplace expansion – and introduce the two concepts of the minor and the cofactor, to help us compute determinants more systematically.

The Minor

When computing a determinant, the first important concept is the minor.

Given an $n \times n$ square matrix $\mathbf{A}$:

$$\mathbf{A} = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n} \end{bmatrix}$$

Matrix $\mathbf{A}$'s element a_{ij} in row i and column j has a corresponding minor denoted M_{ij} .

The minor M_{ij} is the determinant of the minor matrix that remains after removing row i and column j from matrix $\mathbf{A}$

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$$M_{i,j} = \det(\mathbf{A}_{i,j})$$

Here, the minor matrix $\mathbf{A}_{i,j}$ has shape $(n-1) \times (n-1)$.

⚠ Note that the minor matrix $\mathbf{A}_{i,j}$ is a matrix, while the minor $M_{i,j}$ is a scalar.

For example

For the 3×3 square matrix $\mathbf{A}$, every element a_{ij} has a corresponding 2×2 minor matrix, and each minor matrix (a matrix) has its own minor (a scalar).

As shown in Figure 1 (a), the element $a_{1,1}$ in row 1, column 1 has minor

$$M_{1,1} = \begin{vmatrix} a_{2,2} & a_{2,3} \\ a_{3,2} & a_{3,3} \end{vmatrix} = a_{2,2}a_{3,3} - a_{2,3}a_{3,2}$$

Figure 1 (a), the blue square represents the element $a_{1,1}$; the gray (and also blue) cells are the removed elements; the green represents the remaining elements, which make up the minor matrix.

Figure 1 (b), the element $a_{1,2}$ in row 1, column 2 has minor

$$M_{1,2} = \begin{vmatrix} a_{2,1} & a_{2,3} \\ a_{3,1} & a_{3,3} \end{vmatrix} = a_{2,1}a_{3,3} - a_{2,3}a_{3,1}$$

Figure 1 (c), the element $a_{1,3}$ in row 1, column 3 has minor

$$M_{1,3} = \begin{vmatrix} a_{2,1} & a_{2,2} \\ a_{3,1} & a_{3,2} \end{vmatrix} = a_{2,1}a_{3,2} - a_{2,2}a_{3,1}$$

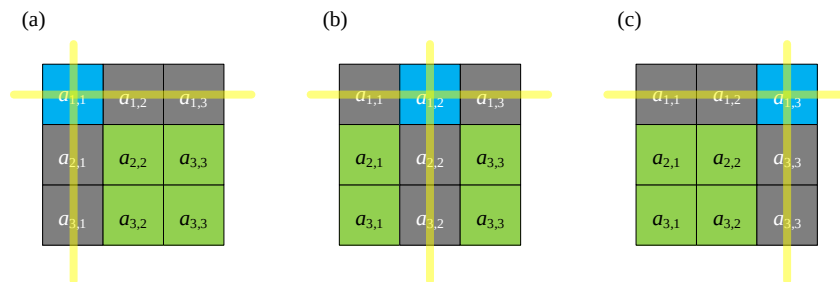


Figure 1. The Minor Matrices for Each Element in Row 1 of a 3×3 Square Matrix

We can choose to expand along different rows or columns.

Figure 2 shows the minor matrix for each element when expanding along row 2.

Figure 2 (a), the element $a_{2,1}$ in row 2, column 1 has minor

$$M_{2,1} = \begin{vmatrix} a_{1,2} & a_{1,3} \\ a_{3,2} & a_{3,3} \end{vmatrix} = a_{1,2}a_{3,3} - a_{1,3}a_{3,2}$$

Figure 2 (b), the element $a_{2,2}$ in row 2, column 2 has minor

$$M_{2,2} = \begin{vmatrix} a_{1,1} & a_{1,3} \\ a_{3,1} & a_{3,3} \end{vmatrix} = a_{1,1}a_{3,3} - a_{1,3}a_{3,1}$$

Figure 2 (c), the element $a_{2,3}$ in row 2, column 3 has minor

$$M_{2,3} = \begin{vmatrix} a_{1,1} & a_{1,2} \\ a_{3,1} & a_{3,2} \end{vmatrix} = a_{1,1}a_{3,2} - a_{1,2}a_{3,1}$$

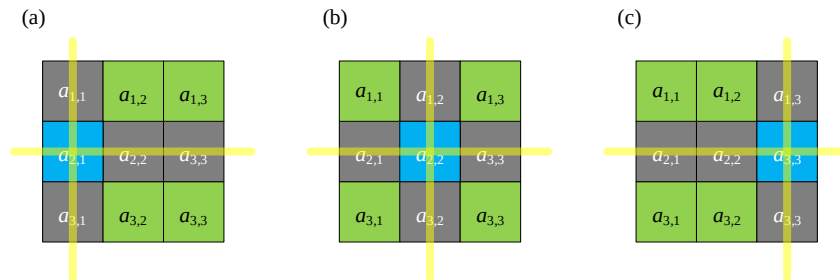


Figure 2. The Minor Matrices for Each Element in Row 2 of a 3×3 Square Matrix

Figure 3 shows the minor matrix for each element when expanding along row 3; please compute these three minors on your own.

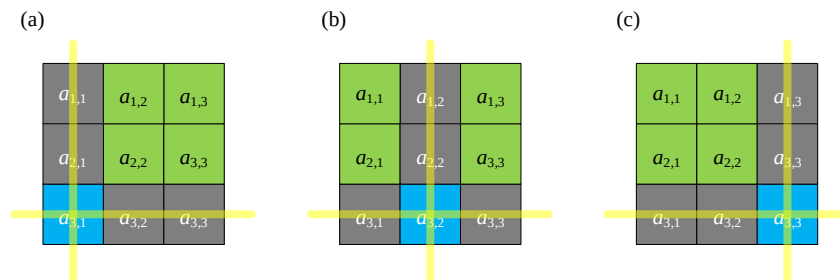


Figure 3. The Minor Matrices for Each Element in Row 3 of a 3×3 Square Matrix

Substituting Specific Values

Given the following 3×3 square matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

According to , the element $a_{1,1}$ in row 1, column 1 has minor

$$M_{1,1} = \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} = (5 \times 9) - (6 \times 8) = 45 - 48 = -3$$

According to , the element $a_{1,2}$ in row 1, column 2 has minor

$$M_{1,2} = \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} = (4 \times 9) - (6 \times 7) = 36 - 42 = -6$$

According to , the element $a_{1,3}$ in row 1, column 3 has minor

$$M_{1,3} = \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} = (4 \times 8) - (5 \times 7) = 32 - 35 = -3$$

Please compute the remaining six minors of the square matrix given in on your own.

The Cofactor

The cofactor introduces a sign adjustment on top of the minor.

The cofactor C_{ij} is defined by the following formula:

$$C_{i,j} = (-1)^{i+j} M_{i,j}$$

Here, $(-1)^{i+j}$ is the factor that adjusts the sign based on the element's position.

Taking the square matrix given in as an example, let's compute the cofactor of the element $a_{1,1}$ in row 1, column 1

$$C_{1,1} = (-1)^{1+1} M_{1,1} = 1 \times (-3) = -3$$

Let's compute the cofactor of the element $a_{1,2}$ in row 1, column 2

$$C_{1,2} = (-1)^{1+2} M_{1,2} = (-1) \times (-6) = 6$$

Let's compute the cofactor of the element $a_{1,3}$ in row 1, column 3

$$C_{1,3} = (-1)^{1+3} M_{1,3} = 1 \times (-3) = -3$$

Since $(-1)^{i+j}$ forms an alternating positive/negative pattern, the sign-adjustment matrix for the cofactors of a 3×3 square matrix is

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

Laplace Expansion

Laplace expansion provides a more systematic method for computing determinants. We can expand along any row or column to solve for the determinant.

For example, expanding along row i to compute $\mathbf{A}$'s determinant,

$$\det(\mathbf{A}) = \sum_{j=1}^n a_{i,j} C_{i,j}$$

For example, if we specify expansion along row 1, the determinant is

$$\det(\mathbf{A}) = \sum_{j=1}^n a_{1,j} C_{1,j}$$

For the 3×3 square matrix $\mathbf{A}$, we can expand along row 1; $\mathbf{A}$'s determinant is

$$\det(\mathbf{A}) = a_{1,1}C_{1,1} + a_{1,2}C_{1,2} + a_{1,3}C_{1,3}$$

Substituting the results computed in ~ :

$$\det(\mathbf{A}) = (1 \times (-3)) + (2 \times 6) + (3 \times (-3)) = -3 + 12 - 9 = 0$$

This also tells us that there are many different paths to solving for a square matrix's determinant.

Steps

Summarizing the above, we can arrive at the specific steps for solving a determinant by hand:

- Select the row or column to use for the determinant expansion, and extract the minor matrices (square matrices).
- Compute the minors (scalars) from the minor matrices (square matrices);
- Compute the cofactor (a scalar) of each element, that is, the sign-adjusted minor (a scalar);
- Using Laplace expansion, multiply each element of the selected row or column by its cofactor, and sum the results to finally obtain the value of the determinant.

Through these steps, we can compute determinants systematically, rather than relying solely on memorized formulas. This method can be extended to higher-dimensional matrices.

A Complete Example

Given the following 3×3 square matrix $\mathbf{A}$, let's follow the steps above to compute the determinant step by step, expanding along rows and columns.

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 0 & 1 \\ 1 & 2 & 1 \end{bmatrix}$$

a) Extract the Minor Matrices

Extract the minor matrix corresponding to each element $\mathbf{A}_{ij}$.

Let's review the concept of the minor matrix. For the 3×3 square matrix $\mathbf{A}$, $\mathbf{A}_{ij}$ is the 2×2 matrix that remains after removing row i and column j .

a.1) If we choose to expand along row 1, the minor matrix $\mathbf{A}_{1,1}$ (with row 1, column 1 removed) is

$$\mathbf{A}_{1,1} = \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix}$$

The minor matrix $\mathbf{A}_{1,2}$ (with row 1, column 2 removed):

$$\mathbf{A}_{1,2} = \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix}$$

The minor matrix $\mathbf{A}_{1,3}$ (with row 1, column 3 removed):

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$$\mathbf{A}_{1,3} = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix}$$

a.2) If we choose to expand along row 2, the minor matrix $\mathbf{A}_{2,1}$ (with row 2, column 1 removed):

$$\mathbf{A}_{2,1} = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}$$

The minor matrix $\mathbf{A}_{2,2}$ (with row 2, column 2 removed):

$$\mathbf{A}_{2,2} = \begin{bmatrix} 1 & 3 \\ 1 & 1 \end{bmatrix}$$

The minor matrix $\mathbf{A}_{2,3}$ (with row 2, column 3 removed):

$$\mathbf{A}_{2,3} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$$

a.3) If we choose to expand along row 3, the minor matrix $\mathbf{A}_{3,1}$ (with row 3, column 1 removed):

$$\mathbf{A}_{3,1} = \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}$$

The minor matrix $\mathbf{A}_{3,2}$ (with row 3, column 2 removed):

$$\mathbf{A}_{3,2} = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$$

The minor matrix $\mathbf{A}_{3,3}$ (with row 3, column 3 removed):

$$\mathbf{A}_{3,3} = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$$

b) Compute the Minors

Compute the determinant of each minor matrix $\mathbf{A}_{ij}$, that is, the minor M_{ij} .

b.1) The minors of each element in row 1. Minor $M_{1,1}$:

$$M_{1,1} = \begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix} = (0 \times 1) - (1 \times 2) = 0 - 2 = -2$$

Minor $M_{1,2}$:

$$M_{1,2} = \begin{vmatrix} 3 & 1 \\ 1 & 1 \end{vmatrix} = (3 \times 1) - (1 \times 1) = 3 - 1 = 2$$

Minor $M_{1,3}$:

$$M_{1,3} = \begin{vmatrix} 3 & 0 \\ 1 & 2 \end{vmatrix} = (3 \times 2) - (0 \times 1) = 6 - 0 = 6$$

b.2) The minors of each element in row 2. Minor $M_{2,1}$:

$$M_{2,1} = \begin{vmatrix} 2 & 3 \\ 2 & 1 \end{vmatrix} = (2 \times 1) - (3 \times 2) = 2 - 6 = -4$$

Minor $M_{2,2}$:

$$M_{2,2} = \begin{vmatrix} 1 & 3 \\ 1 & 1 \end{vmatrix} = (1 \times 1) - (3 \times 1) = 1 - 3 = -2$$

Minor $M_{2,3}$:

$$M_{2,3} = \begin{vmatrix} 1 & 2 \\ 1 & 2 \end{vmatrix} = (1 \times 2) - (2 \times 1) = 2 - 2 = 0$$

b.3) The minors of each element in row 3. Minor $M_{3,1}$:

$$M_{3,1} = \begin{vmatrix} 2 & 3 \\ 0 & 1 \end{vmatrix} = (2 \times 1) - (3 \times 0) = 2 - 0 = 2$$

Minor $M_{3,2}$:

$$M_{3,2} = \begin{vmatrix} 1 & 3 \\ 3 & 1 \end{vmatrix} = (1 \times 1) - (3 \times 3) = 1 - 9 = -8$$

Minor $M_{3,3}$:

$$M_{3,3} = \begin{vmatrix} 1 & 2 \\ 3 & 0 \end{vmatrix} = (1 \times 0) - (2 \times 3) = 0 - 6 = -6$$

c) Compute the Cofactors

The cofactor is simply the minor with a sign added, following the rule $C_{i,j} = (-1)^{i+j} M_{i,j}$.

c.1) The cofactors of each element in row 1. Cofactor $C_{1,1}$:

$$C_{1,1} = (-1)^{1+1} M_{1,1} = (1) \times (-2) = -2$$

Cofactor $C_{1,2}$:

$$C_{1,2} = (-1)^{1+2} M_{1,2} = (-1) \times (2) = -2$$

Cofactor $C_{1,3}$:

$$C_{1,3} = (-1)^{1+3} M_{1,3} = (1) \times (6) = 6$$

c.2) The cofactors of each element in row 2. Cofactor $C_{2,1}$:

$$C_{2,1} = (-1)^{2+1} M_{2,1} = (-1) \times (-4) = 4$$

Cofactor $C_{2,2}$:

$$C_{2,2} = (-1)^{2+2} M_{2,2} = (1) \times (-2) = -2$$

Cofactor $C_{2,3}$:

$$C_{2,3} = (-1)^{2+3} M_{2,3} = (-1) \times (0) = 0$$

c.3) The cofactors of each element in row 3. Cofactor $C_{3,1}$:

$$C_{3,1} = (-1)^{3+1} M_{3,1} = (1) \times 2 = 2$$

Cofactor $C_{3,2}$:

$$C_{3,2} = (-1)^{3+2} M_{3,2} = (-1) \times (-8) = 8$$

Cofactor $C_{3,3}$:

$$C_{3,3} = (-1)^{3+3} M_{3,3} = (1) \times (-6) = -6$$

d) (Row Expansion) Compute the Determinant

I'm sure you've already noticed that we've actually provided three "parallel" paths for computing the determinant of a 3×3 square matrix by expanding along a row.

d.1) Expanding Along Row 1 to Compute the Determinant

$$\det(A) = a_{1,1}C_{1,1} + a_{1,2}C_{1,2} + a_{1,3}C_{1,3}$$

Substituting the values:

$$\det(A) = 1 \times (-2) + 2 \times (-2) + 3 \times 6 = -2 - 4 + 18 = 12$$

d.2) Expanding Along Row 2 to Compute the Determinant

$$\det(A) = a_{2,1}C_{2,1} + a_{2,2}C_{2,2} + a_{2,3}C_{2,3}$$

Substituting the values:

$$\det(A) = 3 \times 4 + 0 \times (-2) + 1 \times 0 = 12 + 0 + 0 = 12$$

The result is clearly consistent with .

d.3) Now Expanding Along Row 3 to Compute the Determinant

$$\det(A) = a_{3,1}C_{3,1} + a_{3,2}C_{3,2} + a_{3,3}C_{3,3}$$

Substituting the values:

$$\det(A) = 1 \times 2 + 2 \times 8 + 1 \times (-6) = 2 + 16 - 6 = 12$$

We find that no matter which row we choose to expand along, we get $\det(A) = 12$.

e) (Column Expansion) Compute the Determinant

Since we can compute the determinant by expanding along a row, we can also expand along a column. Let's look at each one below.

e.1) Expanding Along Column 1 to Compute the Determinant

$$\det(A) = a_{1,1}C_{1,1} + a_{2,1}C_{2,1} + a_{3,1}C_{3,1}$$

Substituting the values:

$$\det(A) = 1 \times (-2) + 3 \times 4 + 1 \times 2 = -2 + 12 + 2 = 12$$

e.2) Expanding Along Column 2 to Compute the Determinant

$$\det(A) = a_{1,2}C_{1,2} + a_{2,2}C_{2,2} + a_{3,2}C_{3,2}$$

Substituting the values:

$$\det(A) = 2 \times (-2) + 0 \times (-2) + 2 \times 8 = -4 + 0 + 16 = 12$$

The result is clearly consistent with .

e.3) Now Expanding Along Column 3 to Compute the Determinant

$$\det(A) = a_{1,3}C_{1,3} + a_{2,3}C_{2,3} + a_{3,3}C_{3,3}$$

Substituting the values:

$$\det(A) = 3 \times 6 + 1 \times 0 + 1 \times (-6) = 18 + 0 - 6 = 12$$

We find that no matter which column we choose to expand along, we get $\det(A) = 12$.

At this point, we've found 6 paths for computing the determinant of the 3×3 square matrix, all giving exactly the same result.

When performing Laplace expansion, you should preferentially choose the row or column containing the most 0s, since the computation for those terms becomes simpler. For example, for the following matrix A , expanding along column 2 to compute the determinant takes the least amount of computation (only one 2×2 determinant needs to be computed).

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

Of course, if you encounter a diagonal square matrix, an upper triangular matrix, or a lower triangular matrix, computing the determinant is even simpler.

Computing the Determinant with Python

Code 1 shows how to compute the determinant using Python. Let's go over the key statements below.

a Defines a function called `cal_minor`; `def` means "define." This function has three inputs: a matrix A , a row index `row_idx`, and a column index `col_idx`. Its job is: from this matrix, delete the specified row and column, and what's left is the "minor matrix" we want.

b First, `numpy.delete()` deletes row `row_idx` from matrix A ; `axis=0` means deleting a "row." The result after deletion is stored in the variable `A_i`. Next, from the result `A_i` of the previous step, column `col_idx` is deleted; `axis=1` means deleting a "column." The result after deletion is stored in `A_ij`, and this is the "minor matrix" we want.

c Defines a new function called `determinant`. This function's goal is: given a matrix A , it computes this matrix's "determinant."

d First, it checks whether the matrix has just one row and one column. If so, it's just a single number, and no computation is needed — the number is returned directly. Then it checks whether the matrix is 2×2 ; if so, its determinant is computed directly using the fixed rule — multiplying the diagonals and subtracting.

e Starts a loop, with `col_idx` counting from 0 to $n-1$. In other words, we process every column of the matrix's first row. Since we're "expanding along the first row," every column needs to take part in the computation.

Then, it calls the `cal_minor` function written earlier, passing in matrix `A`, row 0, and the index of the current column, to get the "minor matrix" obtained by removing this row and this column.

Then it computes the "cofactor." Note that this uses "recursion": we're calling `determinant()` again, from within itself! The program keeps breaking the matrix down into smaller and smaller matrices, until it becomes 2×2 or 1×1 , at which point it stops.

`det += A[0, col] * cofactor` is the key step — this is the Laplace expansion. We take each element of the original matrix's first row, multiply it by its corresponding "cofactor," and add it to `det`. The program computes the determinant of the larger matrix through exactly this layer-by-layer accumulation.

Code 1. Computing the Determinant |  LA_04_04_01.ipynb

```

## 初始化
import numpy as np

## 自定义函数，提取余子矩阵
a
def cal_minor(A, row_idx, col_idx):
b
    A_i = np.delete(A, row_idx, axis=0)
    A_ij = np.delete(A_i, col_idx, axis=1)
    # 去掉指定的行、列
    return A_ij

## 用Laplace展开递归计算行列式
c
def determinant(A):
d
    n = A.shape[0]
    if n == 1:
        return A[0, 0]
    if n == 2:
        return A[0, 0] * A[1, 1] - A[0, 1] * A[1, 0]

    det = 0
e
    for col_idx in range(n):
        minor = cal_minor(A, 0, col_idx) # 沿第一行展开
        cofactor = ((-1) ** col_idx) * determinant(minor)
        # 计算代数余子式
        # 相当于 (-1) ** ((col_idx + 1) + 1) = (-1) ** col_idx
        det += A[0, col_idx] * cofactor # 计算行列式

    return det

## 测试
A = np.array([[1, 2, 3],
              [3, 0, 1],
              [1, 2, 1]])
A_det = determinant(A)

```



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. For the following square matrices, please choose any row or column and compute the minors (scalars) of the corresponding elements. Also, think about which row or column would be the most convenient to expand along.

►
$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

▶
$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

▶
$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Q2. Use this section's Laplace expansion to compute the determinant of the following 2×2 matrix.

▶
$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Q3. Please use column expansion to compute 's determinant.

Q4. Please study the diagonal rule (Sarrus' rule) on your own.

Q5. Please modify Code 1, adding a statement that checks whether the matrix is square, so the determinant computation proceeds only if it is.

Q6. Please modify Code 1, adding a statement that preferentially selects the row or column with the most zeros to expand along when computing the determinant.